

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 1: Data Cleaning Challenge**

**COURSE NAME:** Query Processing for Data Science **COURSE CODE:** DSA0503

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**COURSE OUTCOME COVERED**

**CO3:** Identify and clean inconsistencies in data by handling missing values, duplicates, outliers, formatting issues, and applying techniques like fuzzy matching and regex.

Assessment Tool Weightage: 35%

**Total Marks: 20**

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**Code of Conduct**

I **Anto Chriswin G 192324093** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



**Debugging**

| Criteria | Anomaly Detection<br>(4 Marks) | Cleaning Methods<br>(4 Marks) | Data Transformation<br>(4 Marks) | Analytical Comparison<br>(4 Marks) | Governance Discussion<br>(4 Marks) |
|----------|--------------------------------|-------------------------------|----------------------------------|------------------------------------|------------------------------------|
| Marks    |                                |                               |                                  |                                    |                                    |

**Total Marks Awarded:**

## Questions:

### Question 1: E-Commerce Customer Transactions Dataset

An online retail company has collected customer transaction data from multiple sales channels. The management wants to identify the top-performing customer segments and products. However, the dataset contains several quality issues.

| CustomerID | CustomerName | Gender | Age | ProductCategory | Quantity | UnitPrice | PurchaseDate | City            |
|------------|--------------|--------|-----|-----------------|----------|-----------|--------------|-----------------|
| C101       | Raj Kumar    | Male   | 28  | Electronics     | 2        | 15000 INR | 10-01-2025   | Chennai         |
| C102       | Priya S      | Female | 35  | Fashion         | 3        | 1200 INR  | 11-01-2025   | chennai         |
| C103       | Arun V       | Male   | 150 | Electronics     | 1        | 300 USD   | 12-01-2025   | CHENNAI         |
| C104       | Kavya R      | Female | 0   | Grocery         | -2       | 800 INR   | 15-01-2025   | Coimbatore      |
| C105       | Rahul P      | Male   | 42  | Fashion         | 5        | 2500 INR  | 18-01-2025   | Madurai         |
| C105       | Rahul P      | Male   | 42  | Fashion         | 5        | 2500 INR  | 18-01-2025   | Madurai         |
| NULL       | Sneha M      | Female | 30  | Electronics     | 2        | 450 USD   | 05-02-2025   | Trichy          |
| C107       | Deepak N     | Male   |     | Grocery         | 3        | 600 INR   | 07-02-2025   | Tiruchirappalli |
| C108       | Meena K      | Female | 27  | Fashion         | 1        | 1800 INR  | 2025-13-01   | Salem           |
| C109       | John D       | Male   | 31  | Electronics     | 4        | 250 EUR   | 15-02-2025   | Chennai         |

### Data Issues

- Missing Customer IDs
- Duplicate transactions
- Negative quantities

- Unit prices recorded in different currencies
- Age values such as 0, 150, and blanks
- Inconsistent city names (e.g., Chennai, chennai, CHENNAI)
- Incorrect purchase dates

## Tasks

1. Identify and classify all data quality issues. (4 Marks)
2. Design a cleaning strategy for each issue. (4 Marks)
3. Perform data standardization and transformation. (4 Marks)
4. Analyze how cleaning affects total revenue calculations. (4 Marks)
5. Generate a report discussing business risks if the data were not cleaned (4 Marks)

```

1
2
3 df["City"] = df["City"].str.strip().str.title()
4
5 df["City"] = df["City"].replace({
6     "Trichy": "Tiruchirappalli"
7 })
8
9 USD_TO_INR = 85
10 EUR_TO_INR = 95
11
12
13 def convert_to_inr(price):
14
15     price = str(price)
16
17     if "INR" in price:
18         value = float(
19             price.replace("INR", "").strip()
20         )
21         return value
22
23     elif "USD" in price:
24         value = float(
25             price.replace("USD", "").strip()
26         )
27         return value * USD_TO_INR
28
29     elif "EUR" in price:
30         value = float(
31             price.replace("EUR", "").strip()
32         )
33         return value * EUR_TO_INR
34
35     return np.nan
36
37
38 df["UnitPrice_INR"] = df["UnitPrice"].apply(convert_to_inr)
39 df["PurchaseDate"] = pd.to_datetime(
40     df["PurchaseDate"],
41     dayfirst=True,
42     errors="coerce"
43 )
44
45 df["Revenue"] = (
46     df["Quantity"] *
47     df["UnitPrice_INR"]
48 )
49
50
51 print("\nCLEANED AND STANDARDIZED DATA")
52 print("-----")
53
54 print(
55     df[
56         [
57             "CustomerID",
58             "CustomerName",
59             "Gender",
60             "Age",
61             "ProductCategory",
62             "Quantity",
63             "UnitPrice_INR",
64             "PurchaseDate",
65             "City",
66             "Revenue"
67         ]
68     ]
69 )
70

```

Output :

```
=====
QUESTION 1: IDENTIFY AND CLASSIFY DATA QUALITY ISSUES
=====

1. Missing Customer IDs - COMPLETENESS ISSUE
CustomerID CustomerName Gender Age ProductCategory Quantity UnitPrice \
6      NaN      Sneha M Female 30.0 Electronics 2 450 USD

PurchaseDate City
6 05-02-2025 Trichy

2. Duplicate Transactions - UNIQUENESS ISSUE
CustomerID CustomerName Gender Age ProductCategory Quantity UnitPrice \
4      C105      Rahul P Male 42.0 Fashion 5 2500 INR
5      C105      Rahul P Male 42.0 Fashion 5 2500 INR

PurchaseDate City
4 18-01-2025 Madurai
5 18-01-2025 Madurai

3. Negative Quantities - VALIDITY ISSUE
CustomerID CustomerName Gender Age ProductCategory Quantity UnitPrice \
3      C104      Kavya R Female 0.0 Grocery -2 800 INR

PurchaseDate City
3 15-01-2025 Coimbatore
```

```
=====
QUESTION 2: CLEANING STRATEGY FOR EACH ISSUE
=====

1. Missing Customer ID
-> Check the original source.
-> If the ID cannot be recovered, assign UNKNOWN_001.

2. Duplicate Transactions
-> Identify duplicate rows.
-> Remove exact duplicate transactions.

3. Negative Quantity
-> Check whether it represents a return.
-> If it is an error, remove or correct the transaction.

4. Different Currencies
-> Convert USD and EUR into INR.
-> Use one standard currency for revenue calculation.

5. Invalid Age
-> Ages 0 and 150 are invalid.
-> Replace invalid values with the median valid age.

6. Missing Age
-> Replace missing age using the median valid age.

7. Inconsistent City Names
-> Convert city names to a common format.
-> Example: Chennai, chennai, CHENNAI -> Chennai.
```

```
=====
QUESTION 3: DATA STANDARDIZATION AND TRANSFORMATION
=====

CLEANED AND STANDARDIZED DATA
-----
CustomerID CustomerName Gender Age ProductCategory Quantity \
0      C101      Raj Kumar Male 28.0 Electronics 2
1      C102      Priya S Female 35.0 Fashion 3
2      C103      Arun V Male 30.5 Electronics 1
4      C105      Rahul P Male 42.0 Fashion 5
6 UNKNOWN_001      Sneha M Female 30.0 Electronics 2
7      C107      Deepak N Male 30.5 Grocery 3
8      C108      Meena K Male 27.0 Fashion 1
9      C109      John D Male 31.0 Electronics 4

UnitPrice_INR PurchaseDate City Revenue
0 15000.0 2025-01-10 Chennai 30000.0
1 1200.0 2025-01-11 Chennai 3600.0
2 25500.0 2025-01-12 Chennai 25500.0
4 2500.0 2025-01-18 Madurai 12500.0
6 38250.0 2025-02-05 Tiruchirappalli 76500.0
7 600.0 2025-02-07 Tiruchirappalli 1800.0
8 1800.0 NaT Salem 1800.0
9 23750.0 2025-02-15 Chennai 95000.0
```

=====

QUESTION 4: EFFECT OF CLEANING ON TOTAL REVENUE

=====

RAW REVENUE

-----

Raw Revenue = 62800.0

CLEANED REVENUE

-----

Cleaned Revenue = 246700.0

REVENUE DIFFERENCE

-----

Difference = 183900.0

REVENUE ANALYSIS

-----

The raw revenue is unreliable because:

1. Duplicate transactions increase revenue.
2. Negative quantities produce incorrect revenue.
3. USD and EUR are not directly comparable with INR.
4. Invalid records can affect revenue calculations.
5. Cleaning produces a more reliable revenue value.

=====

QUESTION 5: BUSINESS RISKS IF DATA IS NOT CLEANED

=====

1. INCORRECT REVENUE REPORTING

Duplicate transactions can overstate total revenue.

2. WRONG CUSTOMER SEGMENTATION

Missing and invalid customer information can result in incorrect customer segments.

3. INCORRECT PRODUCT PERFORMANCE

Different currencies can make product revenue comparisons inaccurate.

4. POOR INVENTORY DECISIONS

Incorrect sales data can result in wrong inventory planning.

5. WRONG MARKETING DECISIONS

Incorrect customer and city information can cause marketing campaigns to target the wrong customers.

6. INCORRECT SALES FORECASTING

Invalid dates, quantities and prices can reduce the accuracy of future sales forecasts.

## Question 2: Hospital Patient Records Dataset

A hospital wants to analyze patient admissions and treatment effectiveness. Data were entered manually by different departments, leading to inconsistencies.

| PatientID | Name       | Age | Gender | AdmissionDate | DischargeDate | Diagnosis    | TreatmentCost | DoctorID |
|-----------|------------|-----|--------|---------------|---------------|--------------|---------------|----------|
| P1001     | Ravi Kumar | 45  | M      | 10-01-2025    | 15-01-2025    | Diabetes     | 25000         | D101     |
| P1002     | Priya S    | 38  | F      | 12-01-2025    | 11-01-2025    | Hypertension | 18000         | D102     |
| P1003     | Arun V     | 125 | M      | 13-01-2025    | 20-01-2025    | diabetes     | 22000         | D101     |

|       |             |    |   |            |            |              |       |      |
|-------|-------------|----|---|------------|------------|--------------|-------|------|
| NULL  | Kavya<br>R  | 29 | F | 15-01-2025 | 17-01-2025 | Diabtes      | 15000 | D103 |
| P1005 | Rahul P     | 51 | M | 18-01-2025 | 25-01-2025 | Asthma       | NULL  | D104 |
| P1005 | Rahul P     | 51 | M | 18-01-2025 | 25-01-2025 | Asthma       | NULL  | D104 |
| P1006 | Sneha<br>M  | 34 | F | 15-01-2025 | 18-01-2025 | Hypertension | 12000 | D102 |
| P1007 | Deepak<br>N |    | M | 01-02-2025 | 05-02-2025 | Asthma       | 10000 | D104 |
| P1008 | Meena<br>K  | 26 | F | 10-02-2025 | 15-02-2025 | HYPERTENSION | 13500 | D102 |
| P1009 | John D      | 40 | M | 12-02-2025 | 18-02-2025 | Diabetes     | 20000 | D101 |

### Data Issues

- Missing patient IDs
- Duplicate patient records
- Discharge dates before admission dates
- Null treatment costs
- Inconsistent diagnosis names
- Multiple formats of dates
- Unrealistic patient ages (>120)

### Tasks

1. Detect all inconsistencies and anomalies. (4 Marks)
2. Develop rules for handling missing and invalid values. (4 Marks)
3. Create a process to standardize diagnosis names. (4 Marks)
4. Evaluate how cleaning changes average treatment cost and hospital stay duration. (4 Marks)
5. Recommend data governance measures to prevent future errors. (4 Marks)

```

df = pd.DataFrame(data)

print("\n" + "="*60)
print("QUESTION 1: DETECT INCONSISTENCIES AND ANOMALIES")
print("="*60)

print("\nMissing Patient IDs:")
print(df[df["PatientID"].isnull()])

print("\nDuplicate Records:")
print(df[df.duplicated(keep=False)])

print("\nDischarge Before Admission:")
ad = pd.to_datetime(df["AdmissionDate"], dayfirst=True)
dd = pd.to_datetime(df["DischargeDate"], dayfirst=True)
print(df[dd < ad])

print("\nMissing Treatment Cost:")
print(df[df["TreatmentCost"].isnull()])

print("\nInconsistent Diagnosis:")
print(df[df["Diagnosis"].unique()])

print("\nUnrealistic Ages:")
print(df[df["Age"] > 120])

print("\nMissing Ages:")
print(df[df["Age"].isnull()])

print("\n" + "="*60)
print("QUESTION 2: RULES FOR MISSING AND INVALID VALUES")
print("="*60)

```

```

df["AdmissionDate"] = pd.to_datetime(
    df["AdmissionDate"], dayfirst=True, errors="coerce"
)

df["DischargeDate"] = pd.to_datetime(
    df["DischargeDate"], dayfirst=True, errors="coerce"
)

df["StayDays_Raw"] = (
    df["DischargeDate"] - df["AdmissionDate"]
).dt.days

raw_avg_cost = df["TreatmentCost"].mean()
raw_avg_stay = df["StayDays_Raw"].mean()

df["PatientID"] = df["PatientID"].fillna("UNKNOWN_001")

df.loc[df["Age"] > 120, "Age"] = np.nan
df["Age"] = df["Age"].fillna(df["Age"].median())

df = df.drop_duplicates()

df.loc[df["DischargeDate"] < df["AdmissionDate"], "DischargeDate"] = pd.NaT

df["TreatmentCost"] = df["TreatmentCost"].fillna(
    df.groupby("Diagnosis")["TreatmentCost"].transform("median")
)

df["StayDays"] = (
    df["DischargeDate"] - df["AdmissionDate"]
).dt.days

clean_avg_cost = df["TreatmentCost"].mean()
clean_avg_stay = df["StayDays"].mean()

```

Output:

```

=====
QUESTION 1: DETECT INCONSISTENCIES AND ANOMALIES
=====

Missing Patient IDs:
  PatientID  Name  Age  Gender  AdmissionDate  DischargeDate  Diagnosis  \
3         NaN  Kavya  R    29.0         F    15-01-2025    17-01-2025    Diabtes

  TreatmentCost  DoctorID
3         15000.0        D103

Duplicate Records:
  PatientID  Name  Age  Gender  AdmissionDate  DischargeDate  Diagnosis  \
4      P1005  Rahul  P    51.0         M    18-01-2025    25-01-2025    Asthma
5      P1005  Rahul  P    51.0         M    18-01-2025    25-01-2025    Asthma

  TreatmentCost  DoctorID
4           NaN        D104
5           NaN        D104

Discharge Before Admission:
  PatientID  Name  Age  Gender  AdmissionDate  DischargeDate  Diagnosis  \
1      P1002  Priya  S    38.0         F    12-01-2025    11-01-2025    Hypertension

  TreatmentCost  DoctorID
1         18000.0        D102

```

```

=====
QUESTION 2: RULES FOR MISSING AND INVALID VALUES
=====

1. Missing PatientID -> Assign a temporary ID or retrieve from source.
2. Duplicate records -> Remove duplicate records.
3. Invalid dates -> Verify and correct from hospital records.
4. Missing TreatmentCost -> Replace using diagnosis median cost.
5. Invalid Age (>120) -> Replace using median valid age.
6. Missing Age -> Replace using median valid age.
7. Invalid Diagnosis -> Standardize using approved diagnosis names.

=====
QUESTION 3: STANDARDIZE DIAGNOSIS NAMES
=====
Standardized Diagnoses:
<StringArray>
['Diabetes', 'Hypertension', 'Asthma']
Length: 3, dtype: str

```

```

QUESTION 4: EFFECT OF CLEANING
=====

Before Cleaning:
Average Treatment Cost = 16937.5
Average Hospital Stay = 4.5 days

After Cleaning:
Average Treatment Cost = 16166.67
Average Hospital Stay = 4.88 days

Changes:
Treatment Cost Change = -770.83
Stay Duration Change = 0.38 days

=====
QUESTION 5: DATA GOVERNANCE RECOMMENDATIONS
=====

1. Use mandatory PatientID validation.
2. Prevent duplicate records using unique PatientID rules.
3. Validate admission and discharge dates automatically.
4. Use standard diagnosis codes such as ICD codes.
5. Set valid age ranges during data entry.
6. Make treatment cost fields mandatory.
7. Use a common date format across departments.
8. Perform regular data-quality audits.
9. Provide data-entry training to hospital staff.
10. Maintain role-based access and change logs.

```

### Question 3: Banking Loan Approval Dataset

A bank uses customer data to determine loan eligibility. The dataset contains errors that may affect risk assessment and approval decisions.

| CustomerID | AnnualIncome | CreditScore | LoanAmount | EmploymentStatus | Age | ExistingDebt | LoanStatus |
|------------|--------------|-------------|------------|------------------|-----|--------------|------------|
|------------|--------------|-------------|------------|------------------|-----|--------------|------------|



|      |          |      |         |               |    |        |          |
|------|----------|------|---------|---------------|----|--------|----------|
| L001 | 750000   | 720  | 300000  | Full Time     | 35 | 50000  | Approved |
| L002 | 900K     | 680  | 400000  | FT            | 42 | 100000 | Approved |
| L003 | 1200000  | NULL | 600000  | fulltime      | 28 | 200000 | Approved |
| L004 | 150000   | 450  | -200000 | Part Time     | 30 | 50000  | Rejected |
| L005 | 5000000  | 790  | 1000000 | Self Employed | 16 | 10000  | Approved |
| L006 | 450000   | 350  | 250000  | Unemployed    | 40 | 300000 | Rejected |
| L007 | 75000000 | 800  | 500000  | Full Time     | 45 | 100000 | Approved |
| L008 | 650000   | 690  | 350000  | FT            | 33 | 70000  | Approved |
| L008 | 650000   | 690  | 350000  | FT            | 33 | 70000  | Approved |
| L010 | NULL     | 710  | 300000  | Full Time     | 37 | 80000  | Approved |

### Data Issues

- Missing credit scores
- Income values recorded in different units
- Negative loan amounts
- Duplicate customer IDs
- Age values below 18
- Employment status entered differently (FT, Full Time, fulltime)
- Extreme outliers in income

### Tasks

1. Identify critical issues that could impact loan decisions. (4 Marks)
2. Propose cleaning techniques for missing and inconsistent data. (4 Marks)
3. Detect and justify treatment of outliers. (4 Marks)
4. Analyze the effect of cleaning on loan approval rates. (4 Marks)
5. Discuss ethical implications of incorrect data cleaning decisions. (4 Marks)

```

1 df = pd.DataFrame(data)
2
3 print("\n" + "-"*60)
4 print("QUESTION 1: CRITICAL ISSUES IMPACTING LOAN DECISIONS")
5 print("-"*60)
6
7 print("\nMissing Credit Scores:")
8 print(df[df["CreditScore"].isnull()])
9
10 print("\nDifferent Income Formats:")
11 print(df["AnnualIncome"])
12
13 print("\nNegative Loan Amounts:")
14 print(df[df["LoanAmount"] < 0])
15
16 print("\nDuplicate Customer IDs:")
17 print(df[df.duplicated("CustomerID", keep=False)])
18
19 print("\nInvalid Ages:")
20 print(df[df["Age"] < 18])
21
22 print("\nInconsistent Employment Status:")
23 print(df["EmploymentStatus"].unique())
24
25 print("\nExtreme Income Values:")
26 print(df[df["AnnualIncome"].astype(str).str.replace("x", "000", regex=False).astype(float) > 10000000])
27
28
29 print("\n" + "-"*60)
30 print("QUESTION 2: CLEANING TECHNIQUES")
31 print("-"*60)
32
33 print("""
34 1. Missing CreditScore -> Replace with median credit score.
35 2. Income units -> Convert all income values to INR.
36 3. Negative LoanAmount -> Treat as invalid and remove/correct.
37 4. Duplicate CustomerID -> Remove duplicate records.
38 5. Age below 18 -> Treat as invalid and verify from source.
39 6. EmploymentStatus -> Standardize to full time and fulltime.
40 7. Missing Income -> Replace using median income or verify source.
41 """)
42
43
44 print("\n" + "-"*60)
45 print("QUESTION 3: OUTLIER DETECTION AND TREATMENT")
46 print("-"*60)
47
48 df["Income"] = df["AnnualIncome"].str.replace(
49     "x", "000", regex=False
50 ).astype(float)
51
52 Q1 = df["Income"].quantile(0.25)
53 Q3 = df["Income"].quantile(0.75)
54 IQR = Q3 - Q1
55
56 lower = Q1 - 1.5 * IQR
57 upper = Q3 + 1.5 * IQR
58
59 outliers = df[
60     (df["Income"] < lower) |
61     (df["Income"] > upper)
62 ]
63
64 print("\nIncome Outliers:")
65 print(outliers[["CustomerID", "Income"]])
66
67 print("\nOutlier Treatment:")
68 print("Extreme income values are checked against source records.")
69 print("Verified values are retained; incorrect values are capped or corrected.")
70
71
72 print("\n" + "-"*60)
73 print("QUESTION 4: EFFECT OF CLEANING ON LOAN APPROVAL RATE")
74 print("-"*60)
75
76 raw_approval_rate = (
77     df["LoanStatus"].eq("Approved").mean() * 100
78 )
79
80 df["CustomerID"] = df["CustomerID"].astype(str)
81
82 df["CreditScore"] = df["CreditScore"].fillna(
83     df["CreditScore"].median()
84 )
85
86 df["Income"] = df["Income"].fillna(
87     df["Income"].median()
88 )
89
90 df["EmploymentStatus"] = df["EmploymentStatus"].replace({
91     "part-time": "Full Time",
92     "fulltime": "Full Time"
93 })
94
95 df = df.drop_duplicates("CustomerID")
96
97 df = df[df["LoanAmount"] >= 0]
98
99 df = df[df["Age"] >= 18]
100
101 clean_approval_rate = (
102     df["LoanStatus"].eq("Approved").mean() * 100
103 )
104
105 print("\nBefore Cleaning:")
106 print("Total Customers =", 10)
107 print("Approved Loans =", (df["LoanStatus"] == "Approved").sum())
108 print("Approval Rate =", round(raw_approval_rate, 2), "%")
109
110 print("\nAfter Cleaning:")
111 print("Total Customers =", len(df))
112 print("Approved Loans =", (df["LoanStatus"] == "Approved").sum())
113 print("Approval Rate =", round(clean_approval_rate, 2), "%")
114
115 print("\nChange in Approval Rate:")
116 print(
117     round(clean_approval_rate - raw_approval_rate, 2),
118     "%"
119 )
120
121
122 print("\n" + "-"*60)
123 print("QUESTION 5: ETHICAL IMPLICATIONS")
124 print("-"*60)
125
126 print("""
127 1. Incorrect cleaning may unfairly reject eligible customers.
128 2. Removing valid outliers can discriminate against high-income customers.
129 3. Incorrect credit score imputation can change loan decisions.
130 4. Missing data should not automatically be interpreted as poor credit.
131 5. Cleaning rules must be transparent and consistently applied.
132 6. Customer financial data must be protected and kept confidential.
133 7. Automated loan decisions should be monitored for bias.
134 8. Important decisions should allow human review when necessary.
135 """)
136
137
138 print("\n" + "-"*60)
139 print("FINAL CLEANED DATA")
140 print("-"*60)
141
142 print(df)

```

Output :

```
=====
QUESTION 1: CRITICAL ISSUES IMPACTING LOAN DECISIONS
=====

Missing Credit Scores:
  CustomerID AnnualIncome  CreditScore  LoanAmount EmploymentStatus  Age \
2          L003      1200000          NaN      600000          fulltime    28

  ExistingDebt LoanStatus
2      200000   Approved

Different Income Formats:
0      750000
1       900K
2     1200000
3     150000
4     5000000
5     450000
6     7500000
7     650000
8     650000
9         NaN
Name: AnnualIncome, dtype: str

Negative Loan Amounts:
  CustomerID AnnualIncome  CreditScore  LoanAmount EmploymentStatus  Age \
3          L004      150000      450.0     -200000          Part Time    30
```

```
=====
QUESTION 2: CLEANING TECHNIQUES
=====

1. Missing CreditScore -> Replace with median credit score.
2. Income units -> Convert all income values to INR.
3. Negative LoanAmount -> Treat as invalid and remove/correct.
4. Duplicate CustomerID -> Remove duplicate records.
5. Age below 18 -> Treat as invalid and verify from source.
6. EmploymentStatus -> Standardize FT, Full Time and fulltime.
7. Missing Income -> Replace using median income or verify source.

=====
QUESTION 3: OUTLIER DETECTION AND TREATMENT
=====

Income Outliers:
  CustomerID  Income
4          L005  5000000.0
6          L007  75000000.0

Outlier Treatment:
Extreme income values are checked against source records.
Verified values are retained; incorrect values are capped or corrected.
```

```
=====
QUESTION 4: EFFECT OF CLEANING ON LOAN APPROVAL RATE
=====

Before Cleaning:
Total Customers = 10
Approved Loans = 8
Approval Rate = 80.0 %

After Cleaning:
Total Customers = 7
Approved Loans = 6
Approval Rate = 85.71 %

Change in Approval Rate:
5.71 %

=====
QUESTION 5: ETHICAL IMPLICATIONS
=====

1. Incorrect cleaning may unfairly reject eligible customers.
2. Removing valid outliers can discriminate against high-income customers.
3. Incorrect credit-score imputation can change loan decisions.
4. Missing data should not automatically be interpreted as poor credit.
5. Cleaning rules must be transparent and consistently applied.
6. Customer financial data must be protected and kept confidential.
7. Automated loan decisions should be monitored for bias.
8. Important decisions should allow human review when necessary.
```

Question 4: Smart City Traffic Monitoring Dataset

A smart city project collects traffic sensor data to optimize traffic signals. Sensor malfunctions and transmission errors have affected data quality.

|          |           |              |              |                     |          |                  |
|----------|-----------|--------------|--------------|---------------------|----------|------------------|
| SensorID | Timestamp | VehicleCount | AverageSpeed | TrafficSignalStatus | RoadName | WeatherCondition |
|----------|-----------|--------------|--------------|---------------------|----------|------------------|

|      |                     |     |     |        |                               |        |
|------|---------------------|-----|-----|--------|-------------------------------|--------|
| S101 | 01-01-2025<br>08:00 | 120 | 45  | Green  | Anna Salai                    | Sunny  |
| S102 | 01-01-2025<br>08:05 | 135 | 52  | Red    | anna salai                    | Sunny  |
| S103 | 01-01-2025<br>08:10 | -50 | 48  | Green  | ANNA SALAI                    | Cloudy |
| S104 | 01-01-2025<br>08:15 | 140 | 350 | Yellow | GST Road                      | Rainy  |
| S105 | NULL                | 110 | 42  | Green  | GST ROAD                      | Sunny  |
| S106 | 01-01-2025<br>08:25 | 130 | 46  | Red    | OMR                           | NULL   |
| S106 | 01-01-2025<br>08:25 | 130 | 46  | Red    | OMR                           | NULL   |
| S107 | 01-01-2025<br>08:30 | 150 | 55  | Green  | Old<br>Mahabalipura<br>m Road | Sunny  |
| S10A | 01-01-2025<br>08:35 | 145 | 53  | Green  | OMR                           | Cloudy |
| S109 | 01-01-2025<br>08:40 | 138 | 50  | Green  | GST Road                      | Rainy  |

### Data Issues

- Missing timestamps
- Duplicate sensor readings
- Vehicle counts recorded as negative numbers
- Average speed exceeding 300 km/h
- Missing weather information
- Inconsistent road naming conventions
- Sensor IDs with formatting errors

### Tasks

1. Assess the quality of the traffic dataset. (4 Marks)
2. Recommend cleaning techniques for time-series data. (4 Marks)
3. Design rules to identify faulty sensor readings. (4 Marks)
4. Evaluate how cleaning affects congestion analysis. (4 Marks)
5. Suggest automated monitoring techniques for future data collection. (4 Marks)



## Code :

### QUESTION 1: ASSESS QUALITY OF THE TRAFFIC DATASET

#### Missing Timestamps:

|   | SensorID | Timestamp | VehicleCount | AverageSpeed | TrafficSignalStatus | \ |
|---|----------|-----------|--------------|--------------|---------------------|---|
| 4 | S105     | NaN       | 110          | 42           | Green               |   |

|   | RoadName | WeatherCondition |
|---|----------|------------------|
| 4 | GST ROAD | Sunny            |

#### Duplicate Sensor Readings:

|   | SensorID | Timestamp        | VehicleCount | AverageSpeed | TrafficSignalStatus | \ |
|---|----------|------------------|--------------|--------------|---------------------|---|
| 5 | S106     | 01-01-2025 08:25 | 130          | 46           | Red                 |   |
| 6 | S106     | 01-01-2025 08:25 | 130          | 46           | Red                 |   |

|   | RoadName | WeatherCondition |
|---|----------|------------------|
| 5 | OMR      | NaN              |
| 6 | OMR      | NaN              |

#### Negative Vehicle Counts:

|   | SensorID | Timestamp        | VehicleCount | AverageSpeed | TrafficSignalStatus | \ |
|---|----------|------------------|--------------|--------------|---------------------|---|
| 2 | S103     | 01-01-2025 08:10 | -50          | 48           | Green               |   |

|   | RoadName   | WeatherCondition |
|---|------------|------------------|
| 2 | ANNA SALAI | Cloudy           |

### QUESTION 2: CLEANING TECHNIQUES FOR TIME-SERIES DATA

1. Convert Timestamp into datetime format.
2. Sort records according to Timestamp.
3. Remove duplicate sensor readings.
4. Replace negative vehicle counts with NaN and verify them.
5. Replace unrealistic speeds above 300 km/h with NaN.
6. Fill missing weather using nearby readings or sensor records.
7. Standardize road names.
8. Validate and correct SensorID formats.
9. Use interpolation only for suitable numerical time-series values.

### QUESTION 3: RULES TO IDENTIFY FAULTY SENSOR READINGS

1. VehicleCount < 0 -> Faulty reading.
2. AverageSpeed < 0 -> Faulty reading.
3. AverageSpeed > 300 -> Faulty reading.
4. Missing Timestamp -> Invalid reading.
5. Duplicate SensorID + Timestamp -> Duplicate reading.
6. Invalid SensorID format -> Sensor configuration error.
7. Missing WeatherCondition -> Incomplete reading.
8. Sudden extreme changes from previous readings -> Possible sensor fault.

### QUESTION 4: EFFECT OF CLEANING ON CONGESTION ANALYSIS

#### Before Cleaning:

Average Vehicle Count = 114.8

Average Speed = 78.7 km/h

#### After Cleaning:

Average Vehicle Count = 136.86

Average Speed = 49.86 km/h

#### Change:

Vehicle Count Change = 22.06

Speed Change = -28.84 km/h

Cleaning improves congestion analysis because invalid negative vehicle counts and unrealistic speeds can produce false traffic conditions. Removing duplicate readings also prevents certain time periods from being overrepresented.

### QUESTION 5: AUTOMATED MONITORING TECHNIQUES

1. Real-time range validation for vehicle count and speed.
2. Automatic duplicate detection using SensorID and Timestamp.
3. Automatic timestamp validation.
4. Sensor ID format validation.

### Question 5: Global Sales Performance Dataset

A multinational company has merged sales data from five countries. Management wants to evaluate regional performance and forecast future sales. The merged dataset contains severe data quality problems.

| Region        | Country | SalesPerson | Product             | SalesAmount | Currency | SalesDate  | CustomerRating |
|---------------|---------|-------------|---------------------|-------------|----------|------------|----------------|
| Asia          | India   | Raj Kumar   | Laptop              | 50000       | INR      | 10-01-2025 | 4              |
| Asia          | India   | Rajkumar    | Laptop              | 55000       | INR      | 10-01-2025 | 5              |
| Europe        | Germany | Anna M      | Laptop              | 1200        | EUR      | 12-01-2025 | 4              |
| Europe        | Germany | Anna M      | Laptop              | 1200        | EUR      | 12-01-2025 | 4              |
| North America | USA     | John D      | Laptop              | 1500        | USD      | 01-15-2025 | 6              |
| Asia          | India   | Priya S     | Mobile              | NULL        | INR      | 18-01-2025 | 3              |
| Europe        | France  | Pierre L    | Ordinateur Portable | 1300        | EUR      | 20-01-2025 | 4              |
| Asia          | India   | Rahul P     | Mobile              | 9999999     | INR      | 25-01-2025 | 5              |
| UK            | UK      | David T     | Laptop              | 1100        | GBP      | 28-01-2025 | 2              |
| Asia          | India   | Priya S     | Smartphone          | 35000       | INR      | 30-01-2025 | 0              |

#### Data Issues

- Multiple currencies (USD, EUR, INR, GBP)
- Missing sales amounts
- Duplicate regional entries
- Customer ratings outside the scale (1–5)
- Different date formats across countries
- Product names in multiple languages
- Salespersons recorded with spelling variations
- Extreme sales outliers due to data-entry errors

#### Tasks

1. Develop a comprehensive data-cleaning framework for multinational datasets. (4 Marks)
2. Design methods to standardize currencies, products, and dates. (4 Marks)
3. Handle missing values and outliers using appropriate analytical techniques. (4 Marks)
4. Compare regional sales rankings before and after cleaning. (4 Marks)
5. Prepare an executive report explaining how data quality influences strategic business decisions. (4 Marks)

```

1  #!/usr/bin/env python
2  """
3  Data quality analysis script.
4  """
5
6  # Import necessary libraries
7  import sys
8  import os
9  import pandas as pd
10 import numpy as np
11
12 # Define file paths
13 data_dir = "data"
14 output_dir = "output"
15
16 # Load data
17 data = pd.read_csv(os.path.join(data_dir, "data.csv"))
18
19 # Data quality analysis
20
21 # 1. Missing values
22 print("\n1. Missing values")
23
24 # Check for missing values in each column
25 for column in data.columns:
26     missing_values = data[column].isnull().sum()
27     print(f"Column: {column}, Missing values: {missing_values}")
28
29 # 2. Duplicate rows
30 print("\n2. Duplicate rows")
31
32 # Check for duplicate rows
33 duplicate_rows = data.duplicated()
34 print(f"Number of duplicate rows: {duplicate_rows.sum()}")
35
36 # 3. Outliers
37 print("\n3. Outliers")
38
39 # Check for outliers using the IQR method
40 Q1 = data["value"].quantile(0.25)
41 Q3 = data["value"].quantile(0.75)
42 IQR = Q3 - Q1
43 lower_bound = Q1 - 1.5 * IQR
44 upper_bound = Q3 + 1.5 * IQR
45
46 # Count outliers
47 outliers = data[(data["value"] < lower_bound) | (data["value"] > upper_bound)]
48 print(f"Number of outliers: {outliers.shape[0]}")
49
50 # 4. Data distribution
51 print("\n4. Data distribution")
52
53 # Calculate the mean, median, and standard deviation
54 mean = data["value"].mean()
55 median = data["value"].median()
56 std_dev = data["value"].std()
57
58 # Print the statistics
59 print(f"Mean: {mean}, Median: {median}, Standard Deviation: {std_dev}")
60
61 # 5. Data quality score
62 print("\n5. Data quality score")
63
64 # Calculate the data quality score based on the above metrics
65 score = 100 - (missing_values.sum() * 10 + duplicate_rows.sum() * 20 + outliers.shape[0] * 10)
66
67 # Print the score
68 print(f>Data quality score: {score}")
69
70 # 6. Data quality report
71 print("\n6. Data quality report")
72
73 # Generate a report summarizing the data quality analysis
74 report = f"""
75 Data Quality Report
76
77 1. Missing values: {missing_values.sum()}
78 2. Duplicate rows: {duplicate_rows.sum()}
79 3. Outliers: {outliers.shape[0]}
80 4. Data distribution: Mean={mean}, Median={median}, Std Dev={std_dev}
81 5. Data quality score: {score}
82
83 """
84
85 # Save the report to a file
86 with open(os.path.join(output_dir, "report.txt"), "w") as f:
87     f.write(report)
88
89 # Print the report
90 print(report)
91
92 # 7. Data quality visualization
93 print("\n7. Data quality visualization")
94
95 # Create a bar chart showing the number of missing values for each column
96 missing_values_counts = data.isnull().sum()
97
98 # Plot the bar chart
99 missing_values_counts.plot(kind="bar")
100
101 # Save the plot to a file
102 plot_path = os.path.join(output_dir, "missing_values.png")
103 missing_values_counts.to_image(plot_path)
104
105 # Print the plot path
106 print(f>Plot saved to: {plot_path}")
107
108 # 8. Data quality conclusion
109 print("\n8. Data quality conclusion")
110
111 # Summarize the findings and provide recommendations
112 conclusion = f"""
113 Data quality conclusion:
114
115 The data quality analysis revealed several issues, including missing values, duplicate rows, and outliers. The data quality score is {score}, indicating a need for improvement. Recommendations include:
116
117 1. Remove missing values or impute them.
118 2. Remove duplicate rows.
119 3. Investigate outliers and determine if they are valid data points.
120 4. Implement data validation rules to prevent future data quality issues.
121
122 """
123
124 # Print the conclusion
125 print(conclusion)
126
127 # 9. Data quality monitoring
128 print("\n9. Data quality monitoring")
129
130 # Set up a monitoring system to track data quality over time
131 # This is a placeholder for a more complex monitoring system
132
133 # 10. Data quality summary
134 print("\n10. Data quality summary")
135
136 # Provide a final summary of the data quality analysis
137 summary = f"""
138 Data quality summary:
139
140 The data quality analysis was completed successfully. The data quality score is {score}, and the report has been saved to {report_path}. The plot showing missing values has been saved to {plot_path}.
141
142 """
143
144 # Print the summary
145 print(summary)
146
147 # End of script
148

```



Output :

```
=====
QUESTION 1: COMPREHENSIVE DATA-CLEANING FRAMEWORK
=====

Missing Sales Amount:
  Region Country SalesPerson Product SalesAmount Currency SalesDate \
5  Asia  India  Priya S  Mobile  NaN  INR  18-01-2025

  CustomerRating
5  3

Duplicate Entries:
  Region Country SalesPerson Product SalesAmount Currency SalesDate \
2  Europe  Germany  Anna M  Laptop  1200.0  EUR  12-01-2025
3  Europe  Germany  Anna M  Laptop  1200.0  EUR  12-01-2025

  CustomerRating
2  4
3  4

Invalid Customer Ratings:
  Region Country SalesPerson Product SalesAmount Currency \
4  North America  USA  John D  Laptop  1500.0  USD
9  Asia  India  Priya S  Smartphone  35000.0  INR
```

```
=====
QUESTION 2: STANDARDIZE CURRENCIES, PRODUCTS AND DATES
=====

Currency Standardization:
  SalesAmount Currency Sales_INR
0  50000.0  INR  50000.0
1  55000.0  INR  55000.0
2  1200.0  EUR  110400.0
3  1200.0  EUR  110400.0
4  1500.0  USD  127500.0
5  NaN  INR  NaN
6  1300.0  EUR  119600.0
7  9999999.0  INR  9999999.0
8  1100.0  GBP  118800.0
9  35000.0  INR  35000.0

Standardized Products:
<StringArray>
['Laptop', 'Mobile']
Length: 2, dtype: str

Standardized Salespersons:
<StringArray>
['Raj Kumar', 'Anna M', 'John D', 'Priya S', 'Pierre L', 'Rahul P', 'David T']
Length: 7, dtype: str
```

```
=====
QUESTION 3: MISSING VALUES AND OUTLIERS
=====
```

Detected Sales Outliers:

|   | SalesPerson | Sales_INR |
|---|-------------|-----------|
| 5 | Priya S     | 5017499.5 |
| 7 | Rahul P     | 9999999.0 |

Missing Sales Amounts:

Filled using product-wise median.

Customer Ratings:

Invalid ratings were replaced using the median rating.

Outliers:

Extreme values were capped using the IQR upper limit.

```
=====
QUESTION 4: REGIONAL SALES RANKING BEFORE AND AFTER CLEANING
=====
```

BEFORE CLEANING:

| Region        |            |
|---------------|------------|
| Asia          | 10139999.0 |
| Europe        | 340400.0   |
| North America | 127500.0   |
| UK            | 118800.0   |

Name: RawSales\_INR, dtype: float64

```
=====
QUESTION 5: EXECUTIVE REPORT
=====
```

EXECUTIVE REPORT

Data quality has a direct effect on strategic business decisions.

1. Revenue Accuracy  
Incorrect currencies, duplicates and outliers can produce incorrect total sales and regional revenue.
2. Regional Performance  
Poor-quality data can make one region appear stronger or weaker than it actually is.
3. Sales Forecasting  
Extreme sales values and missing values can reduce the accuracy of future sales forecasts.
4. Customer Analysis  
Invalid customer ratings can lead to incorrect conclusions about customer satisfaction.
5. Product Strategy  
Different product names can split the same product into multiple categories and distort product performance.

**Signature of the Course Faculty**